TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REGISTER

Version 1.1 – proposed final version for Review 2 (25/10/2026). Changes from version 1.0 (28/06/2026) are listed in the attached Record of Changes (Form BM03).

Class: **Duration time**: from 09/2026 To 03/2027

(*) Profession: Software Engineer **Specialty**: SE X

(*) Kinds of person make registers: Lecturer x Students

1. Register information for supervisor (if have)

No.	Full name	Phone	**E-Mail**	**Title**
Supervisor	Thân Thị Ngọc Vân	0912656836	vanttn@fpt.edu.vn	

2. Register information for students (if have)

	Full name	**Student code**	**Phone**	**E-mail**	**Role in Group**
1	Nguyễn Phương Mỹ Thuận	SE171113	0377168254	thuannpmse171113@fpt.edu.vn	Leader
2	Lê Đức Trung Thi	DE180553	0905749864	thildtde180553@fpt.edu.vn	Member
3	Huỳnh Quốc Việt	SE182482	0813076315	viethqse182482@fpt.edu.vn	Member
4	Phan Huỳnh Hải Phượng		0708639363	haifuong2408@gmail.com	Member
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3. Register content of Capstone Project

(*) 3.1. Capstone Project name:

English: Develop an AI-Assisted Omnichannel Order Fulfillment and Packaging Optimization System

Vietnamese: Xây dựng hệ thống hỗ trợ xử lý đơn hàng đa kênh và tối ưu hóa đóng gói thông minh bằng trí tuệ nhân tạo

a. Context:

The rapid growth of e-commerce has enabled businesses to sell products through multiple online channels such as Facebook Marketplace, TikTok Shop, Shopee, Lazada, and company-owned

websites. While omnichannel selling increases business opportunities, it also introduces significant operational challenges. Orders from different platforms are managed independently, forcing warehouse staff to switch between multiple systems, manually consolidate orders, verify inventory, prepare packages, print shipping labels, and arrange deliveries. These repetitive manual processes increase processing time, labor costs, packaging errors, and shipping expenses.

Packaging is another critical challenge in order fulfillment. Many sellers rely on personal experience when selecting package sizes, cushioning materials, and shipping methods. Inappropriate packaging may increase shipping costs, waste packaging materials, reduce transportation efficiency, or even damage products during delivery. Furthermore, the growing demand for sustainable packaging requires businesses to optimize both packaging costs and environmental impact.

Recent advances in Artificial Intelligence, optimization algorithms, and logistics analytics provide new opportunities to automate warehouse operations. AI can intelligently group orders, recommend optimal packing strategies, calculate package dimensions and weight, select suitable packaging materials, estimate shipping costs, and recommend the most appropriate shipping carrier. Combined with real-time order synchronization across multiple e-commerce platforms, such technologies can significantly improve fulfillment efficiency while reducing operational costs.

This project proposes the development of an AI-assisted Omnichannel Order Fulfillment and Intelligent Packaging Optimization System. The system will synchronize orders from multiple sales channels, manage warehouse fulfillment workflows, optimize packaging decisions using AI, estimate logistics costs, generate shipping labels and packing slips, monitor fulfillment progress, and provide analytical dashboards to support data-driven operational decisions.

You’re a business selling on multiple platforms like Facebook, TikTok, Shopee, etc. Now you need a system to optimize your order packing process by consolidating all orders from these platforms to optimize your business. This project proposes the development of an AI-assisted Omnichannel Order Fulfillment and Intelligent Packaging Optimization System. The system will synchronize orders from multiple sales channels, manage warehouse fulfillment workflows, optimize packaging decisions using AI, estimate logistics costs, generate shipping labels and packing slips, monitor fulfillment progress, and provide analytical dashboards to support data-driven operational decisions.

Functional requirement :

The system shall synchronize orders automatically from:

● Lazada (read-only)
● Company Website – AURELLE, built by the team (Open API + webhooks)

The synchronization process shall include:

● New orders
● Order updates
● Order cancellations
● Recipient information
● Product catalog
● AURELLE return requests
● Write-back to AURELLE: order / return status and available stock

Order Consolidation

The system shall:

- Merge orders from different platforms
- Group orders by customer
- Group orders by destination
- Group orders by warehouse
- Detect duplicate orders
- Prioritize urgent orders

Warehouse Fulfillment

Support:

- Picking List Generation
- Batch Picking
- Order Verification
- Packing Workflow
- Fulfillment Tracking
- Packing Status Monitoring

AI-assisted Packaging Optimization

The AI engine shall recommend:

- Appropriate carton size
- Number of cartons required
- Package dimensions
- Estimated package weight
- Suitable cushioning materials
- Packing sequence
- Empty space reduction
- Fragile item protection

Optimization objectives include:

- Minimize packaging cost
- Minimize shipping cost
- Reduce unused space
- Protect products
- Improve sustainability

Shipping Cost Estimation

Estimate:

- Package weight
- Volumetric weight
- Shipping fee
- Packaging cost
- Total fulfillment cost

Recommend:

- Suitable courier
- Delivery method
- Expected delivery time

Label & Document Generation

Generate:

● Packing Slip
● Shipping Label
● Barcode
● QR Code
● Delivery Manifest

Support batch printing.

Fulfillment Tracking

Track:

● Picking
● Packing
● Shipping
● Delivery
● Returns

Provide real-time status updates.

Dashboard & Analytics

Display:

● Packing reports (summary and feedback)
● Packaging-material savings
● Action history with CSV export (up to 5,000 rows)

Non-functional requirement:

Performance

The system shall synchronize orders from multiple e-commerce platforms in near real time and support high-volume order processing with minimal latency.

Scalability

The platform shall support additional marketplaces, warehouses, shipping carriers, and AI optimization services without requiring significant architectural modifications.

Security

The system shall implement secure authentication, role-based access control, encrypted API communication, audit logging, and secure storage of customer and transaction data.

(*) 3.2. Main proposal content (including result and product)

Proposed Solutions:

Develop an AI-Assisted Omnichannel Order Fulfillment and Packaging Optimization System.

System Actors

1. Store Owner

● View the connection status of the e-commerce platforms (shops are connected by the System Administrator)
● View consolidated orders
● Monitor fulfillment progress
● View business dashboards
● Configure packaging rules
● Configure shipping preferences
● Analyze operational reports

2. Warehouse Staff

● Receive picking lists

● Pick products
● Pack orders
● Confirm package completion
● Print packing slips
● Print shipping labels
● Update fulfillment status

3. Packaging Staff
● Review AI packaging recommendations
● Select packaging materials
● Confirm package dimensions
● Measure package weight
● Approve packing plans

4. Shipping Coordinator
● Select shipping carrier
● Generate shipping labels
● Schedule shipment pickup
● Track shipment status
● Estimate shipping costs

5. AI Recommendation Engine
● Optimize package size
● Recommend packaging boxes
● Recommend cushioning materials
● Estimate shipping costs
● Optimize packing sequences
● Consolidate multiple orders
● Detect abnormal packages

6. System Administrator
● Manage users
● Manage warehouse configuration
● Manage packaging templates
● Configure AI parameters
● Manage integrations (connect / disconnect marketplace shops)

Applied Theory:

Students are expected to study and apply knowledge from the following areas:

E-commerce Systems
● Omnichannel Commerce
● Order Fulfillment
● Warehouse Management Systems (WMS)
● Inventory Management

Logistics & Supply Chain
● Packaging Engineering
● Shipping Cost Calculation
● Warehouse Operations
● Courier Integration

Artificial Intelligence
● 3D Bin Packing Algorithms
● Optimization Algorithms
● Recommendation Systems
● Machine Learning
● Cost Optimization
Software Engineering
● Distributed Systems
● RESTful APIs
● Event-Driven Architecture
● Microservices
● Integration Patterns
Data Analytics
● Business Intelligence
● Logistics Analytics
● Dashboard Design
● KPI Monitoring
Suggested Technologies
● NestJS (Node.js) – back end
● ReactJS – staff portal; Next.js – company website AURELLE
● Flutter – mobile application (planned in Iteration 3)
● MongoDB
● Redis
● Scheduled jobs, webhooks and transactional outbox (instead of a message broker)
● Docker
● Groq API (OpenAI-compatible LLM API)
● OR-Tools (Google) for optimization – Python packer service

Products (Expected Deliverables):

The final product shall consist of:

Web Application
● Omnichannel Order Management Portal
● Warehouse Fulfillment Dashboard
● AI Packaging Optimization Module
● Shipping Management Module
● Analytics Dashboard
● Configuration Module
Mobile Application
● Warehouse Picking
● Packing Confirmation
● Barcode/QR Scanning
● Shipment Tracking
● Push Notifications
AI Services
● Packaging Recommendation Engine

● Box Size Optimization Engine
● Shipping Recommendation Engine
● Cost Optimization Engine
● Order Consolidation Engine

Integration Services

● Marketplace Integration APIs(at least two chanels)
● Shipping Carrier APIs (optional)
● Barcode & QR Code Service
● Label Printing Service
● Notification Service

Proposed Tasks:

package 1: document that includes design the overall software architecture, database schema, marketplace integration layer, fulfillment workflow engine, REST APIs, AI modules, dashboards, and security mechanisms.

package 2: Omnichannel Integration Module that develops connectors for Lazada and the company website AURELLE to synchronize new orders, order updates and cancellations, recipient information, product catalog and AURELLE return requests, and writes order / return status and available stock back to AURELLE.

package 3: AI Packaging Optimization Module that implements AI-assisted packaging recommendations, box size optimization, packaging material selection, shipping cost estimation, and order consolidation algorithms.

package 4: Fulfillment Management Module that develops warehouse picking, packing, shipping management, label generation, barcode support, fulfillment tracking, and notification services.

package 5: Analytics and Reporting that develops packing reports (summary and feedback), packaging-material savings and the action history with CSV export.

4. Other comments (propose all relative things if have).

Proposed version 1.1 for Review 2. Every change from version 1.0 is listed with its reason in the attached Form BM03 (Record of Changes).

Supervisor (If have) *(Sign and full name)*	HCM, date 28/06/2026 **On behalf of Registers** *(Sign and full name)*

Registration Form – Record of Changes

1. Project Information

Project code	FA26SE036
Project name	Develop an AI-Assisted Omnichannel Order Fulfillment and Packaging Optimization System (OptiPack AI)
Team	OptiPack AI – Nguyễn Phương Mỹ Thuận (leader), Lê Đức Trung Thi, Huỳnh Quốc Việt, Phan Huỳnh Hải Phượng
Supervisor	Thân Thị Ngọc Vân
Current registration version	1.0 (registration form dated 28/06/2026)
Proposed version	1.1
Version	1.0
Date	25/10/2026

2. Changes

☐ No changes to the current registration form

No.	Section	Current content	Proposed content	Reason	Proposed by
1	3.1 Functional requirement – Order synchronization ; 3.2 Proposed Tasks – package 2	Synchronize orders from Facebook Marketplace, TikTok Shop, Shopee, Lazada and Company Website (optional): new orders, order updates, cancellations, customer information, product information, payment status and shipping status.	Order synchronization (new orders, order updates and cancellations, recipient, product catalog) for Lazada and for the team-built company website AURELLE (Open API + webhooks); AURELLE return requests are synchronized and order / return status and available stock are written back to AURELLE; Lazada is read-only. TikTok Shop, Shopee and Facebook Marketplace: removed (adapter files for TikTok Shop and Tiki remain in the code but are not registered).	Shopee: developer account for individual sellers requires a Preferred or Mall Seller shop, which a new shop cannot obtain. Facebook Marketplace: no public API. TikTok Shop: Partner Center requires a business license and a company operating for at least one year. Constraint from partner APIs; the "at least two channels" deliverable is met by Lazada + AURELLE.	Team
2	3.1 Dashboard & Analytics; 3.2 Proposed Tasks – package 5	Dashboards and reports on sales channels, fulfilment, shipping, packaging cost and warehouse efficiency; export to Excel, PDF and CSV.	Packing reports (summary and feedback), packaging-material savings and the action history with CSV export (up to 5,000 rows). Dashboards and reports on sales channels and shipping performance, and Excel / PDF export, are out of scope; home-dashboard widgets for revenue, finance and staff currently show sample data.	Iterations 1–2 concentrated on the fulfilment workflow (integration, warehouse, packing, shipping, returns); the back end has no aggregate API for channel and shipping reports yet.	Team
3	3.2 Suggested technologies	Spring Boot, PostgreSQL, Apache Kafka (suggested).	NestJS (Node.js) instead of Spring Boot, MongoDB instead of PostgreSQL, scheduled jobs + webhooks + transactional outbox instead of Apache Kafka. As suggested: ReactJS staff portal, Redis, Docker, OR-Tools (Python packer service) and an LLM API (Groq); plus a Next.js website. Flutter is kept for the mobile app	The registration form lists these items under "Suggested Technologies". One TypeScript stack (NestJS, React, Next.js) for back end and front ends, document-shaped order data in MongoDB, and the current order	Team

No.	Section	Current content	Proposed content	Reason	Proposed by
			planned in Iteration 3; Elasticsearch (optional) is not used.	volume does not need a message broker.	
4	3.2 System Actors – Store Owner, System Administrator	Store Owner: connect e-commerce platforms.	The System Administrator connects and disconnects marketplace shops (OAuth); the Store Owner views the connection status of the shops. Other Store Owner responsibilities are unchanged.	Connecting a shop gives OptiPack an OAuth token with access to the seller account; the team restricted this sensitive action to the Admin role.	Team

3. Decision at Review 2 (completed by reviewers)

No.	Decision (Accepted / Adjusted / Rejected)	Reviewer note